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- (b) Use kinetic theory of an ideal gas to explain the increase of pressure inside the basketball when air is pumped into it. (2 marks)

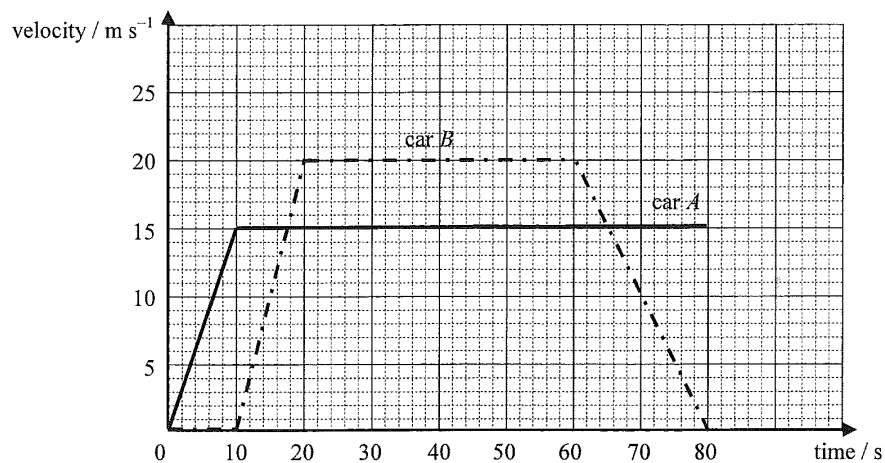
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3. Two cars *A* and *B* initially at the same position, start to travel along the same straight horizontal road. The graph below shows how their velocities vary with time.



- (a) Describe the motion of car *A* along the whole journey from $t = 0$ to $t = 80$ s. (2 marks)

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- (b) (i) Which car attained the greatest acceleration throughout the journey? Find this acceleration. (2 marks)

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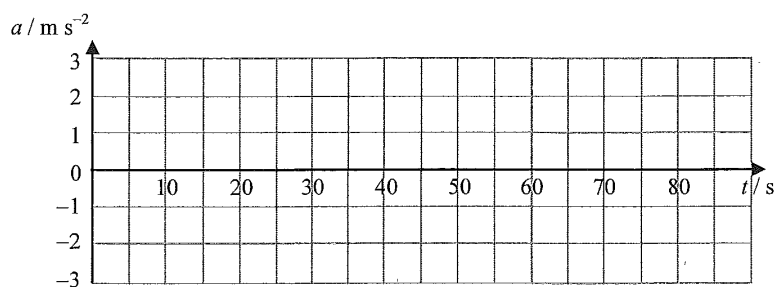
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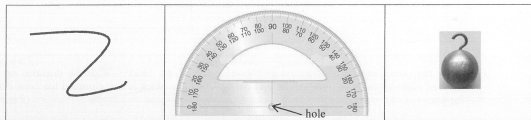
- (ii) Sketch the acceleration-time ($a-t$) graph of car *B* from $t = 0$ to $t = 80$ s. (2 marks)



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5. You are given a long light string, a protractor and a metal ball with a hook.



Suppose you are inside a train which is at rest initially and later it travels along a straight horizontal track with constant acceleration. With the aid of a diagram, describe how to measure the acceleration of the train. Show your working including mathematical derivation. (6 marks)

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6. Read the following description about a **mirage** and answer the questions that follow.

A mirage is often seen on highways during hot summers. Pools of water seem to cover the roadway far ahead. Distant objects appear to be reflected by the surface of the 'water'. The phenomenon is caused by the difference in refractive index between the hot air near the road surface and the cooler air above it. The refractive index of cool air is greater than that of hot air, but the differences are so small that the subsequent deviations of light rays are tiny. Sufficiently large temperature differences between the hot air near the road surface and the above cooler air over a short height (i.e. high temperature gradient) and light rays travelling along sufficiently long path lengths are required to form a mirage.



Figure 6.1 Mirage seen on a highway. This photo was taken with a telephoto lens which gives the perception that the viewer is very close to the car ahead.

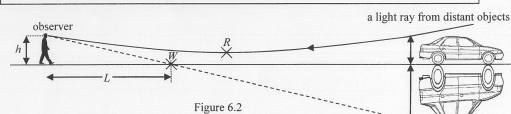


Figure 6.2

Figures 6.2 and 6.3 illustrate the principle of the phenomenon. Air of different temperatures is simplified to several layers and modeled as parallel slabs as shown in Figure 6.3. The bending of the light ray from distant objects is much exaggerated.

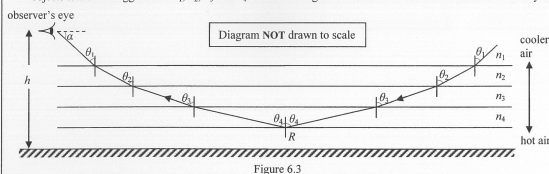


Figure 6.3

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(a) State ONE essential condition for a mirage to be observed.

(1 mark)

(b) (i) Referring to Figure 6.3, deduce the relationship between θ_1 , θ_4 and refractive indices n_1 , n_4 . For total internal reflection just to occur at R , θ_4 can be taken as 90° . Hence, find the corresponding value of θ_1 if $n_1 = 1.000261$ and $n_4 = 1.000221$.

(3 marks)

(ii) Find L in Figure 6.2 if $h = 1.5$ m. (Note: $\alpha + \theta_1 = 90^\circ$ in Figure 6.3.)

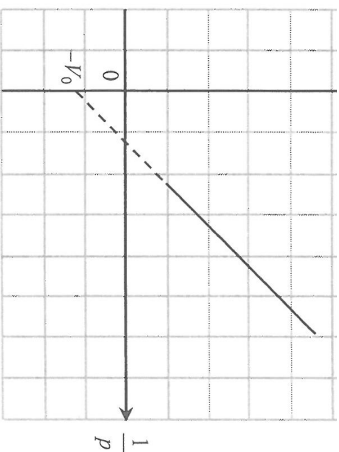
(2 marks)

(c) A thirsty traveller in a vast desert sees similar mirages such that a 'water source' appears at W which is distance L away like the one in Figure 6.2. If he walks a distance L towards that 'water source', how far would the 'water source' appear to him? Explain your answer.

(2 marks)

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Figure 2.2



(i) State ONE experimental precaution for keeping the gas temperature constant.

(1 mark)

(ii) The straight line graph does not pass through the origin but cuts the vertical axis at $-V_0$ instead. Suggest what V_0 stands for.

(1 mark)

(iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure 2.2. (2 marks)

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- (a) Calculate the acceleration of the lift from $t = 0$ to $t = 2$ s. (2 marks)

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The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 395 N.

- (b) Match these readings with the three stages, A , B and C , of the ride (shown in Figure 3.1). Hence deduce the mass of the person. (3 marks)

A : _____ B : _____ C : _____

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- (iii) Given that the area of the water surface is 0.014 m^2 . Estimate the **increase** in the force acting on the water due to the trapped air. (2 marks)

- (c) Given that the mass of the plastic bottle is negligible.

- (i) After the rocket-launching control is pulled and a jet of water ejects from the nozzle, explain why the rocket can go upwards using Newton's laws of motion. (2 marks)

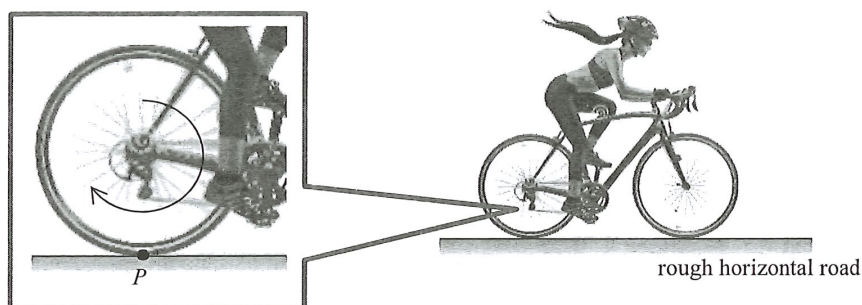
- (ii) Just after pulling the control, what is **most** of the energy stored in the pressurized air converted to? (1 mark)

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3. Figure 3.1 shows a cyclist riding on a bicycle travelling with a constant velocity of 8.0 m s^{-1} on a rough horizontal road.

Figure 3.1



- (a) The rear wheel is rotating clockwise without slipping on the ground as shown and P is the contact point with the road at that moment. Indicate in the figure the frictional force acting on the rear wheel at P . (1 mark)
- (b) If the effective resistive force experienced by the cyclist and the bicycle is 17.0 N when riding at this constant velocity, estimate the mechanical power delivered by the cyclist. (2 marks)

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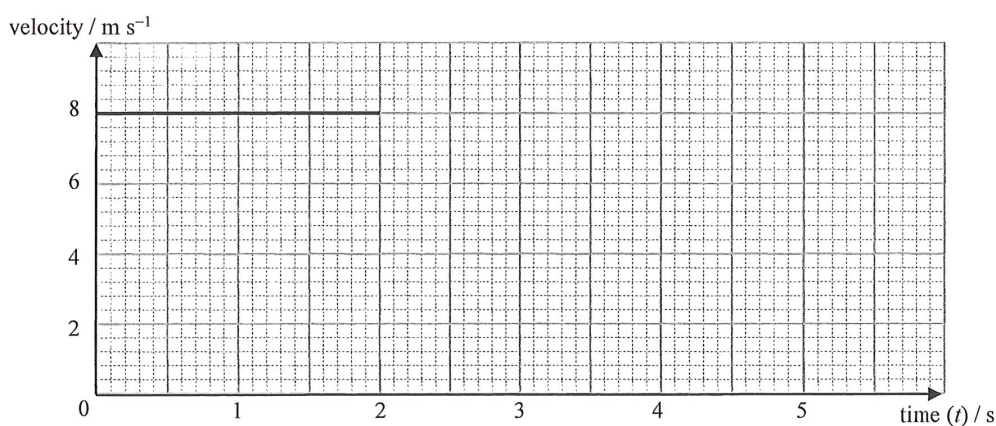
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- (c) At time $t = 2.0 \text{ s}$, the cyclist sees a small obstacle 9 m directly in front of the bicycle and applies the brakes as soon as possible. The reaction time of the cyclist is 0.2 s . The bicycle then decelerates uniformly and finally stops at $t = 3.8 \text{ s}$.

- (i) Complete the velocity-time graph below from $t = 2.0 \text{ s}$ to $t = 3.8 \text{ s}$. (2 marks)



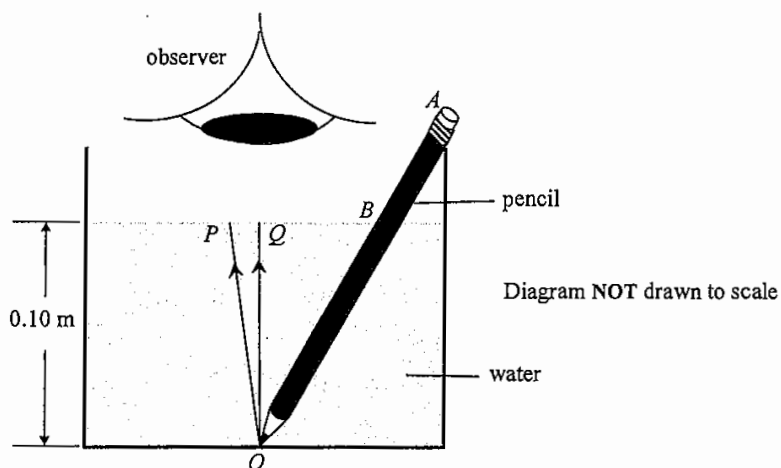
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8. Figure 8.1 shows a pencil in a glass of water with an observer viewing vertically downwards. Given that the refractive index of water is 1.33.

Figure 8.1



Light rays P and Q come from the tip O of the pencil, with Q along the vertical.

- (a) On Figure 8.1, complete the paths of these light rays and indicate the image of O seen by the observer (label this as I). (2 marks)
- (b) The angle between ray P and ray Q is 1.5° . Find the angle of refraction of ray P . (2 marks)

- (c) Hence, or otherwise, find the distance between image I and the water surface if the depth of water is 0.10 m. (2 marks)

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2. A trolley is moving on a horizontal surface with a constant velocity of 2 m s^{-1} . At time $t = 0 \text{ s}$, a small bead of mass 0.01 kg is projected upwards with a vertical speed of 5 m s^{-1} from a device P mounted on the trolley as shown. Neglect air resistance.

Figure 2.1



- *(a) Find the time taken, t , for the bead to travel back to the level of projection. (2 marks)

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- (b) What is the kinetic energy of the bead at its maximum height? (2 marks)

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- *(c) When the bead falls back to the level of projection, it is just above device P as shown. Explain why the bead does not fall behind or beyond device P at that moment. (2 marks)

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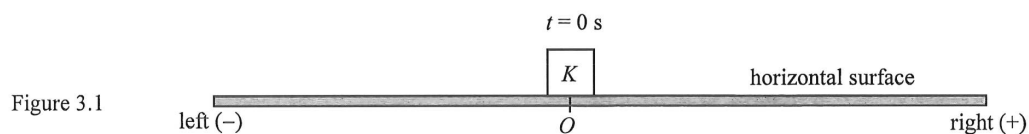
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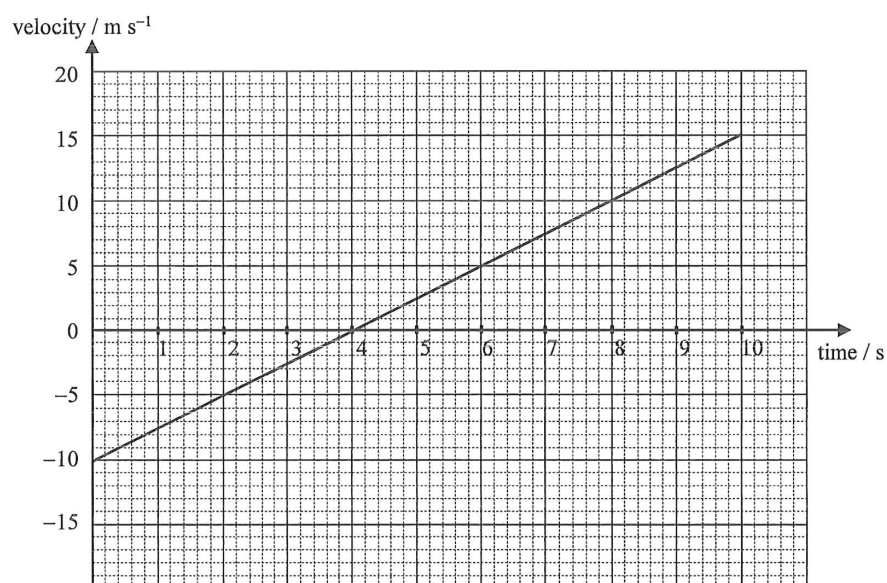
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3.



An object K moving on a horizontal surface is acted upon by a horizontal constant net force (not drawn on the diagram) such that it passes point O at time $t = 0$ s as shown in Figure 3.1. Its velocity-time graph is shown below, taking to the right as positive (+).



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Figure 10.3(a)

Figure 10.3(b)

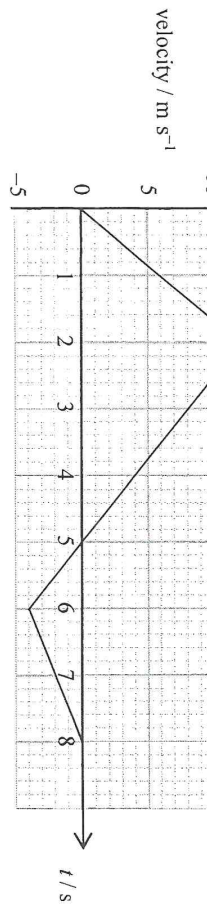
- (i) Given that the refractive index of water for the infra-red is 1.26, find the critical angle of the glass-water interface for the infra-red ray. (3 marks)

- (ii) Sketch in Figure 10.3(b) how the infra-red ray propagates after being incident on the glass-water interface. (2 marks)

- (iii) The wipers are activated when the intensity of the signal received by the infra-red sensor changes. Explain briefly how the intensity changes. (2 marks)

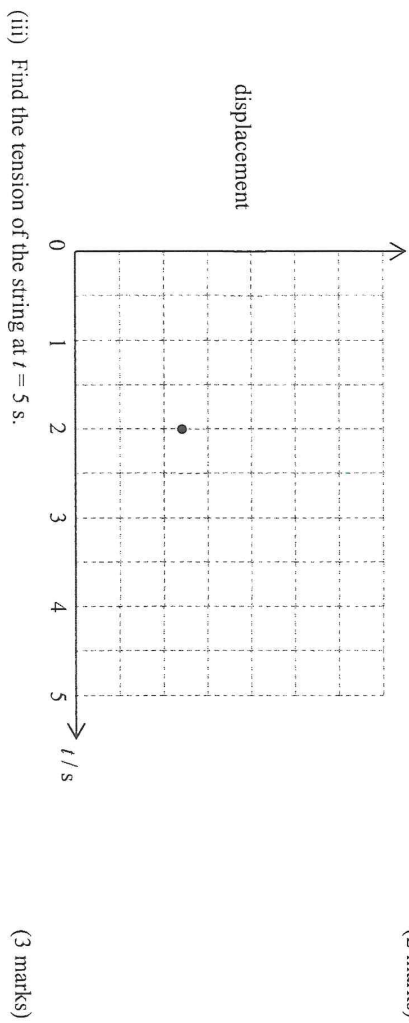
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- (i) Find the maximum height that the block can reach from $t = 0$ to 8 s. (2 marks)

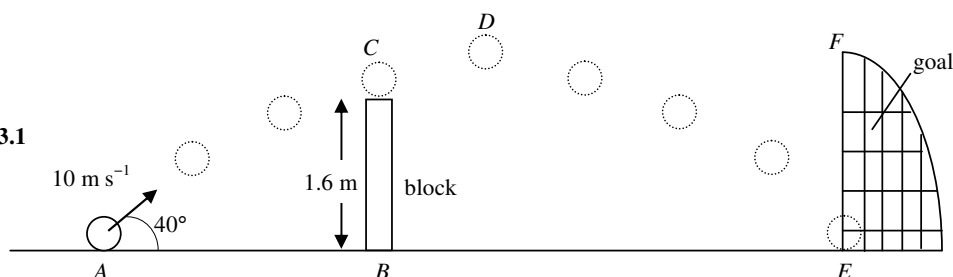
- (ii) Sketch the displacement-time graph of the block from $t = 0$ to 5 s. The displacement at $t = 2$ s is marked. Labelling of values is not required and the displacement at $t = 0$ s is taken to be zero. (2 marks)



- (iii) Find the tension of the string at $t = 5$ s. (3 marks)

3. A ball is kicked and moves with an initial velocity of 10 m s^{-1} at an angle of 40° to the horizontal. The ball then just passes a block of height 1.6 m , reaching the highest point D , and finally hits the ground at E as shown in Figure 3.1. (Neglect air resistance and the size of the ball.)

Figure 3.1



- (a) Draw an arrow to indicate the direction of acceleration of the ball at C . (1 mark)
- *(b) For a projectile of initial velocity u that makes an angle θ with the horizontal, show that its horizontal range is given by $\frac{u^2 \sin 2\theta}{g}$. Hence, or otherwise, find another angle of projection such that the ball can still reach E with the same initial speed of 10 m s^{-1} . (4 marks)
- (Given: $2 \sin \theta \cos \theta = \sin 2\theta$)

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- (c) Calculate the speed of the ball at C . (2 marks)

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- (b) Calculate the magnetic field B through coil C when there is a leakage current of 0.5 A from the load to the Earth. The magnetic field B due to a current-carrying conductor is 1500 times larger in soft iron. (2 marks)

- (c) Electrical appliances are usually equipped with fuses. When a short circuit occurs between the live and neutral wires, the fuse blows but the earth leakage circuit breaker does not operate. Explain these observations. (2 marks)

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9.



Figure 9.1

Figure 9.1 shows a microwave oven. Mary wants to estimate the useful output power of the oven. She is provided with the apparatus and material shown in Figure 9.2.

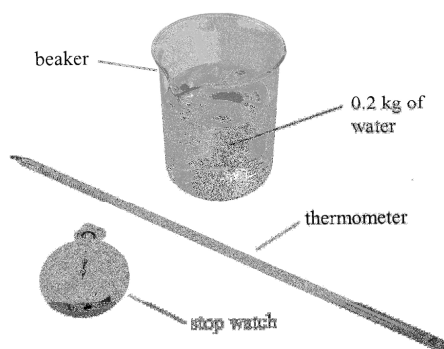


Figure 9.2

- (a) Describe how Mary should conduct the experiment. Specify all measurements Mary has to take. State **EITHER** one precaution taken **OR** one assumption made when conducting this experiment. Write down an equation for calculating the useful output power. (5 marks)

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- (b) The value obtained by Mary is found to be smaller than the rated power of the oven. Suggest one possible reason to account for this difference. (1 mark)

- (c) Explain whether increasing the mass of water used in the experiment would improve the accuracy of the experiment. (1 mark)

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